

Kartikeya Gaur

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EDUCATION

Indian Institute of Science (IISc) <i>M.Tech. in Smart Mobility and Logistics Systems</i>	Bengaluru, India 2027
National Institute of Technology (NIT) <i>B.Tech. in Civil Engineering</i>	Hamirpur, India 2024

EXPERIENCE

Research Intern <i>University of New South Wales (UNSW)</i>	Jun – Jul 2026 <i>Sydney, Australia</i>
Developed a comparative route-choice modeling framework for Sydney and Bengaluru public transit networks using real GTFS and ridership data, implementing Expanded Path Size Logit (EPS) and Nested Recursive Logit (NRL). [Link]	
Junior Research Fellow <i>Indian Institute of Remote Sensing (IIRS), ISRO</i>	Sept 2024 – July 2025 <i>Dehradun, India</i>
Developed an automated framework for river discharge and bathymetry estimation integrating multi-mission satellite altimetry (ICESat-2, SWOT, Sentinel-3), achieving 2.40 m RMSE. [Link]	
Geospatial Project Trainee <i>BISAG-N</i>	Dec 2022 - Jan 2023 <i>Gandhinagar, India</i>
Proposed three railway alignments under the PM GatiShakti National Master Plan and analyzed terrain characteristics and topology in QGIS. [Link]	
Technical Trainee <i>Defence Laboratory (DRDO)</i>	Jun – Jul 2022 <i>Jodhpur, India</i>
Evaluated military trafficability by integrating supervised LULC classifications, DEM derivatives, and digitized soil maps in ArcGIS. [Link]	

PROJECTS

Last-Mile Connectivity Analysis and Visualization <i>Python, Power BI, QGIS</i>
Analyzed large-scale commuter survey data across Bengaluru, Delhi, and Nagpur; mapped Delhi's high non-walk dependency zones (13.35% area, 38/290 wards) and identified Nagpur's maximum population accessibility per station (0.88%). [Link]
Nighttime Street Light Audit Using Computer Vision <i>YOLOv5, YOLOv8, TensorFlow</i>
Deployed an edge-ready YOLOv5/YOLOv8 pipeline trained on a custom 2.1k-frame dataset; optimized inference using FP16/INT8 quantization, pruning, and knowledge distillation. [Link]

TECHNICAL SKILLS

Languages: Python, C++, R, Julia
Geospatial: ArcGIS, QGIS, Geographic Information Systems, Google Earth Engine
Data & Analytics: Pandas, NumPy, SciPy
Scalable Computing: Apache Spark*, Parallel & Distributed Computing*
Tools & BI: Power BI, Jupyter Notebook, Git

PUBLICATIONS

- Gaur, K., et al. (2025). "SWOT Mission with Wide-Swath Altimetry: Observations and Insights into India's Inland Waterbodies." *Journal of the Indian Society of Remote Sensing*.

ACHIEVEMENTS

- Secured All India Rank (AIR) 10:** in GATE 2023 (Geomatics Engineering) with an 819 score. [\[Link\]](#)

* Currently learning through DS221: Introduction to Scalable Systems, IISc.